THE NOVA RESEARCH CURRICULUM

General Science and Social Studies

Grade 5 · Term 1

Student edition

General Science: SC-5.1 vertebrates, SC-5.2 microorganisms, SC-5.3 flowers and seeds. Social Studies: SS-5.1 rights and responsibilities, SS-5.2 government and federating units.

General Science · lesson notes

- **SC-5.1 (Weeks 1–4):** five vertebrate groups: fish (gills, scales), amphibians (moist skin, live in water and on land), reptiles (dry scales, lay eggs on land), birds (feathers, beaks), mammals (hair, feed young on milk). Sort 20 animals.
- **SC-5.2 (Weeks 5–8):** microorganisms: bacteria, viruses, fungi. Useful: yogurt, bread, decomposition. Harmful: diseases, food spoilage. Prevention: handwashing, cooking food well, vaccination.
- **SC-5.3 (Weeks 9–12):** parts of a flower; pollination by insects and wind (self and cross); seed formation; dispersal by wind, water, animals and bursting pods.

Social Studies · lesson notes

- **SS-5.1 (Weeks 1–4):** civic rights (education, equality before law, freedom of expression) and responsibilities (obeying the law, paying taxes as adults, voting at 18); human rights for every person; responsible online behaviour.
- **SS-5.2 (Weeks 5–12):** Pakistan's Constitution of 1973; legislature (Parliament: National Assembly and Senate) makes laws; executive (Prime Minister and cabinet; the President is head of state) carries them out; judiciary (Supreme Court and other courts) settles disputes. Federating units: Punjab, Sindh, Khyber Pakhtunkhwa, Balochistan and Islamabad Capital Territory; Gilgit-Baltistan and Azad Jammu and Kashmir have their own arrangements.

▶ **Try it on NovaSchool** (open **Simulations** on your dashboard):

- **Pollination and seeds** — Send the bee to collect pollen from one flower and carry it to another. Then watch a seed form.

Workbook pages

Page	Outcome	Activity
1	SC-5.1	Vertebrate sorting table.
2	SC-5.2	Useful vs harmful microorganisms poster.
3	SC-5.3	Label a flower; seed-dispersal matching.
4	SS-5.1	Rights and responsibilities pairs.
5	SS-5.2	Three branches diagram; map of federating units.

Question bank

#	Outcome	Question	Your answer
1	SC-5.1	A frog is a(n) ...	reptile / amphibian / fish / mammal
2	SC-5.1	A whale is a mammal. True or false.	True / False
3	SC-5.1	Which group has feathers?	reptiles / birds / fish / mammals
4	SC-5.2	Which microorganism helps make yogurt?	virus / bacteria / mould on bread / algae
5	SC-5.2	All microorganisms are harmful. True or false.	True / False
6	SC-5.2	Give one way to stop harmful germs spreading.	_____
7	SC-5.3	Moving pollen from one flower to another is ...	dispersal / pollination / germination / photosynthesis
8	SC-5.3	Dandelion seeds are spread mainly by ...	water / wind / animals / bursting
9	SS-5.1	Pakistani citizens can vote at the age of ...	16 / 18 / 21 / 25
10	SS-5.1	Name one right and one responsibility of a citizen.	_____
11	SS-5.2	Which branch makes laws?	executive / legislature / judiciary / army
12	SS-5.2	The Supreme Court is part of the ...	legislature / executive / judiciary / Senate
13	SS-5.2	Pakistan's current Constitution was made in ...	1947 / 1956 / 1973 / 1999
14	SS-5.2	Parliament has two houses: the National Assembly and the Senate. True or false.	True / False